

Perbandingan Model Pembelajaran Mesin untuk Prediksi Kanker Payudara

Logistic Regression, K-Nearest Neighbor, Naïve Bayes, Decision Tree

Disusun Oleh Kelompok 1

Kelompok 1

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Pendahuluan



Tujuan Tugas

Membandingkan empat algoritma klasifikasi dalam prediksi kanker payudara.



Dataset

Breast Cancer Wisconsin (Diagnostic) memiliki 569 sampel dan 30 fitur.



Variabel Target

Diagnosis (0 = Malignant, 1 = Benign)

Dataset

Sumber Data

<https://www.kaggle.com/datasets/reihanenamdari/breast-cancer/data>

Statistik

Sebagian besar data berasal dari kelas benign (62.7%) dibandingkan malignant (37.3%).

Deskripsi Dataset

- 569 sampel dengan 30 fitur numerik
- Fitur utama: radius, tekstur, perimeter, area
- Setiap fitur memiliki tiga nilai: mean, standard error, worst

Eksplorasi Data dan Pra-pemrosesan



Menangani Missing Values

Dataset tidak memiliki missing values.



Analisis Korelasi

Beberapa fitur memiliki korelasi tinggi dengan diagnosis.



Feature Selection

Fitur berkorelasi >0.92 dihapus untuk menghindari multikolinearitas.



Normalisasi Data

Data dinormalisasi menggunakan StandardScaler.

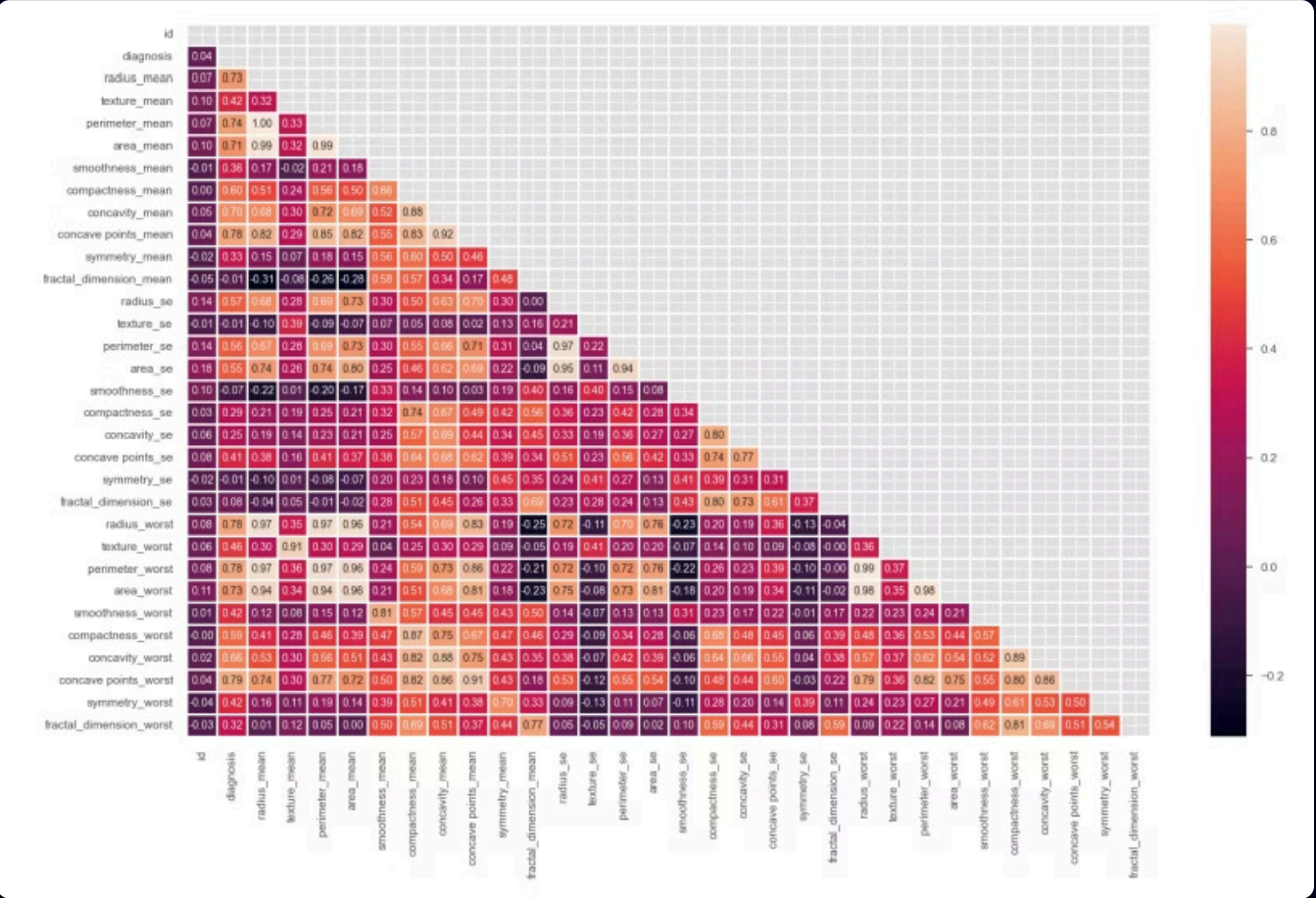
```
df.isnull().sum()
```

```
id          0
diagnosis   0
radius_mean 0
texture_mean 0
perimeter_mean 0
area_mean   0
smoothness_mean 0
compactness_mean 0
concavity_mean 0
concave points_mean 0
symmetry_mean 0
fractal_dimension_mean 0
radius_se    0
texture_se    0
perimeter_se  0
area_se       0
smoothness_se 0
compactness_se 0
concavity_se  0
concave points_se 0
symmetry_se   0
fractal_dimension_se 0
radius_worst  0
texture_worst 0
perimeter_worst 0
...
concavity_worst 0
concave points_worst 0
symmetry_worst  0
fractal_dimension_worst 0
dtype: int64
```

Eksplorasi Data dan Pra-pemrosesan - 1 -

Dataset Tidak memiliki missing value

Eksplorasi Data dan Pra-pemrosesan - 2 -



Analisis Korelasi

```
# feature selection
corr_matrix = df.corr().abs()
mask = np.triu(np.ones_like(corr_matrix, dtype=bool))
tri_df = corr_matrix.mask(mask)

to_drop = [x for x in tri_df.columns if any(tri_df[x]>0.92)]

df = df.drop(to_drop, axis=1)

print(df.shape[1])
```

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Feature Selection

```
# scaling data
from sklearn.preprocessing import StandardScaler
scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)
```

Normalisasi atau Standarisasi Data

Implementasi Model

1. Logistic Regression

Algoritma klasifikasi linear untuk memprediksi probabilitas kejadian variabel kategoris.

2. K-Nearest Neighbors

Mengklasifikasikan data berdasarkan 5 tetangga terdekat menggunakan metrik jarak.

3. Naïve Bayes

Algoritma klasifikasi probabilistik berdasarkan Teorema Bayes dengan asumsi independensi fitur.

4. Decision Tree

Model prediktif menggunakan serangkaian keputusan berbasis aturan.

Implementasi & Evaluasi Model - 1 -

```
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report
print(accuracy_score(y_train, log_reg.predict(X_train)))
log_reg_acc = accuracy_score(y_test, log_reg.predict(X_test))
print(log_reg_acc)
y_pred = log_reg.predict(X_test)
print(confusion_matrix(y_test, y_pred))
print(classification_report(y_test, y_pred))
```

0.989010989010989

0.9649122807017544

[[66 1]

[3 44]]

	precision	recall	f1-score	support
0	0.96	0.99	0.97	67
1	0.98	0.94	0.96	47
accuracy			0.96	114
macro avg	0.97	0.96	0.96	114
weighted avg	0.97	0.96	0.96	114

Logistic Regression

```
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report
print(accuracy_score(y_train, knn.predict(X_train)))
knn_acc = accuracy_score(y_test, knn.predict(X_test))
print(knn_acc)
y_pred = knn.predict(X_test)
print(confusion_matrix(y_test, y_pred))
print(classification_report(y_test, y_pred))
```

0.967032967032967

0.956140350877193

[[66 1]

[4 43]]

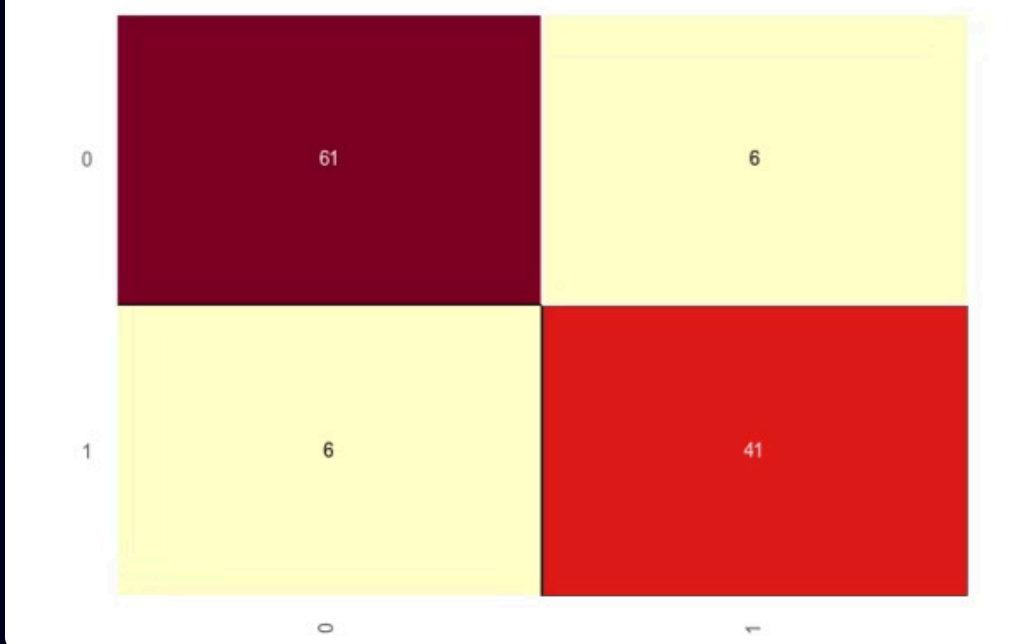
	precision	recall	f1-score	support
0	0.94	0.99	0.96	67
1	0.98	0.91	0.95	47
accuracy			0.96	114
macro avg	0.96	0.95	0.95	114
weighted avg	0.96	0.96	0.96	114

K-Nearest Neighbors

Implementasi & Evaluasi Model - 2 -

```
from yellowbrick.classifier import ConfusionMatrix
cm = ConfusionMatrix(naive_bayes)
cm.fit(X_train, y_train)
cm.score(X_test, y_test)
```

	precision	recall	f1-score	support
0	0.91	0.91	0.91	67
1	0.87	0.87	0.87	47
accuracy			0.89	114
macro avg	0.89	0.89	0.89	114
weighted avg	0.89	0.89	0.89	114



Naive Bayes

```
from sklearn.tree import DecisionTreeClassifier
from sklearn.model_selection import GridSearchCV

dtc = DecisionTreeClassifier()

parameters = {
    'criterion':['gini','entropy'],
    'max_depth':range(2,32,1),
    'min_samples_leaf':range(1,10,1),
    'min_samples_split':range(2,10,1),
    'splitter':['best','random']
}

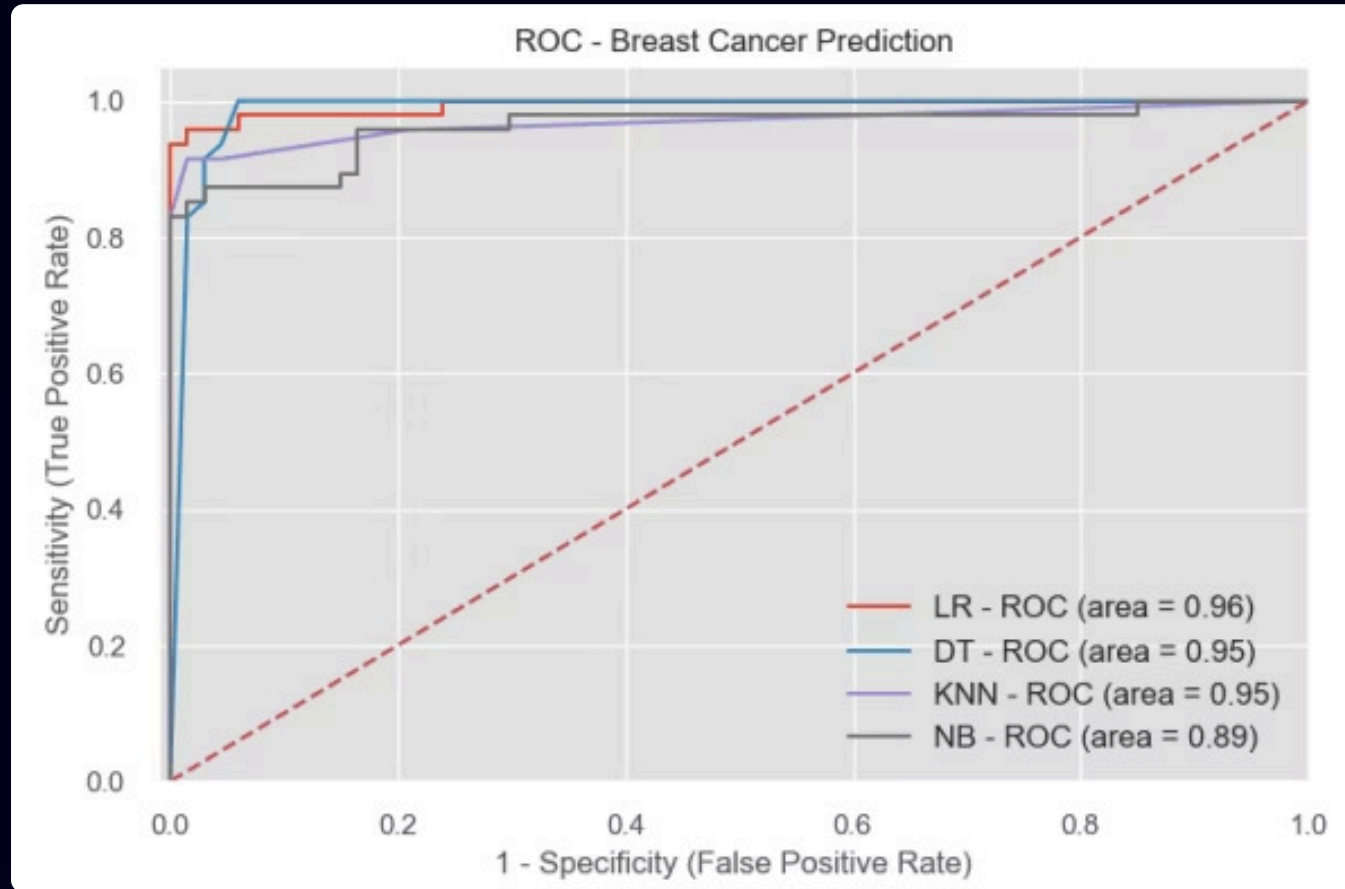
grid_search_dt = GridSearchCV(dtc, parameters, cv=5, n_jobs=-1, verbose=1)
grid_search_dt.fit(X_train, y_train)
```

Decision Tree

```
0.9736263736263736
0.9385964912280702
[[62  5]
 [ 2 45]]
```

	precision	recall	f1-score	support
0	0.97	0.93	0.95	67
1	0.90	0.96	0.93	47
accuracy			0.94	114
macro avg	0.93	0.94	0.94	114
weighted avg	0.94	0.94	0.94	114

Grafik ROC

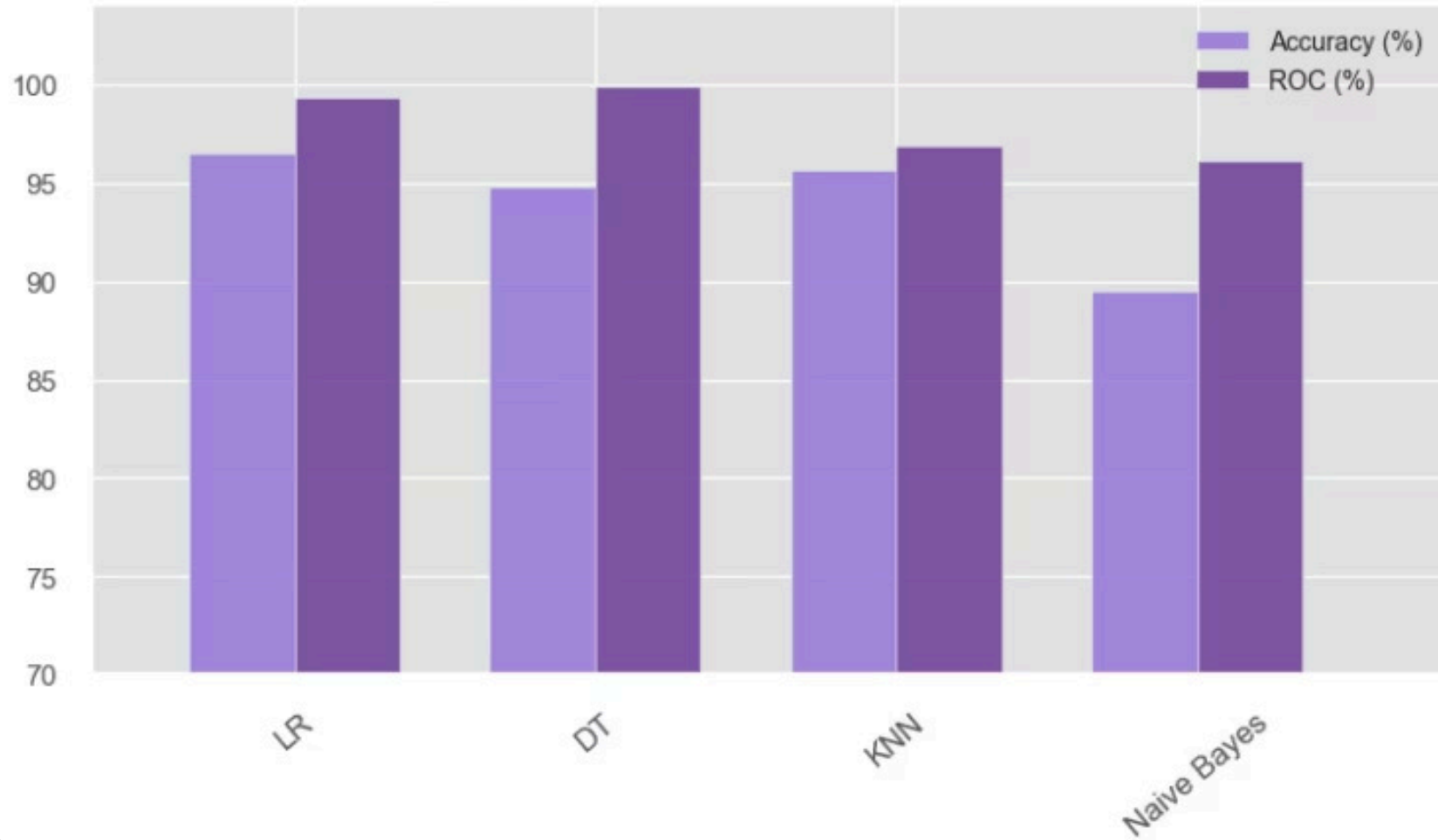


Evaluasi Model - 1 -

Model	Akurasi	ROC-AUC	Precision	Recall	F1-Score
Logistic Regression	96.4%	0.96	0.97	0.96	0.96
KNN	95.6%	0.95	0.95	0.95	0.95
Naïve Bayes	89.0%	0.89	0.89	0.87	0.88
Decision Tree	93.8%	0.95	0.94	0.96	0.95

Tabel ini menunjukkan perbandingan metriks evaluasi dari keempat model.

Performance Evaluation - Breast Cancer Prediction



Evaluasi Model - 2 -

96.4%

Logistic Regression

ROC-AUC: 0.96

95.6%

KNN

ROC-AUC: 0.95

89%

Naïve Bayes

ROC-AUC: 0.89

93.8%

Decision Tree

ROC-AUC: 0.95

Perbandingan Kinerja Model



Logistic Regression

Akurasi 96.4%, ROC-AUC 0.96.

Model terbaik secara keseluruhan.



Decision Tree

Akurasi 93.8%, ROC-AUC 0.95.

Unggul dalam recall kelas positif.



KNN

Akurasi 95.6%, ROC-AUC 0.95.

Kinerja stabil dan setara Decision Tree.



Naïve Bayes

Akurasi 89%, ROC-AUC 0.89.

Performa paling rendah di semua metrik.

Kesimpulan



Rekomendasi

Logistic Regression

Cocok untuk aplikasi prediksi kanker payudara dengan prioritas keseimbangan akurasi.

Decision Tree

Alternatif baik jika fokus pada deteksi kasus positif dengan recall tinggi.

Naïve Bayes

Gunakan dengan hati-hati mengingat performa yang lebih rendah.